Status	Finished
Started	Monday, 30 March 2026, 2:20 PM
Completed	Monday, 30 March 2026, 2:37 PM
Duration	17 mins 16 secs

Question 1

Complete

Marked out of 1.00

In a self-driving car, route selection and path planning belong mainly to which layer?

Select one:

- a. Action
- b. Perception
- c. Learning
- d. Reasoning

Question 2

Complete

Marked out of 1.00

If det(A) ≠ 0 for a 3×3 matrix, what is rank?

Select one:

- a. 1
- b. 2
- c. 3
- d. 0

Question 3

Complete

Marked out of 1.00

If matrix:

$$\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

Eigenvalues are:

Select one:

- a. 2, 3
- b. 1, 2
- c. 0, 0
- d. 5, 6

Question 4

Complete

Marked out of 1.00

Find derivative of $f(x) = 3x^2 + 2x$

Select one:

- a. $6x + 2$
- b. $3x + 2$
- c. $6x$
- d. x^2

Question 5

Complete

Marked out of 1.00

Which statement best describes Prompt Engineering, RAG, Fine-Tuning, and Training from Scratch?

Select one:

- a. They are separate model types
- b. They are forms of data cleaning
- c. They offer different control levels
- d. They replace neural networks

Question 6

Complete

Marked out of 1.00

Find determinant of:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Select one:

- a. -2
- b. 2
- c. 0
- d. 1

Question 7

Complete

Marked out of 1.00

Who described AI as “the science and engineering of making intelligent machines”?

Select one:

- a. Alan Turing
- b. Stuart Russell
- c. John McCarthy
- d. Marvin Minsky

Question 8

Complete

Marked out of 1.00

Find dot product of vectors (1,2,3) and (4,5,6)

Select one:

- a. 32
- b. 20
- c. 15
- d. 10

Question 9

Complete

Marked out of 1.00

A university AI tutor answers students using only approved notes and PDFs. Which design is most likely being used?

Select one:

- a. CNN with OCR
- b. Rule engine with search
- c. Transformer with RAG
- d. Classifier with RL

Question 10

Complete

Marked out of 1.00

What does RAG stand for in AI?

Select one:

- a. Retrieval-Augmented Generation
- b. Recursive AI Graph
- c. Reinforced Adaptive Generation
- d. Retrieval-Aware Grounding

Question 11

Complete

Marked out of 1.00

Which architecture is most closely associated with modern LLMs such as GPT and BERT?

Select one:

- a. Transformer
- b. Decision Tree
- c. Naive Bayes
- d. Random Forest

Question 12

Complete

Marked out of 1.00

Number of non-zero rows in echelon form is 2. Rank = ?

Select one:

- a. 1
- b. 2
- c. 3
- d. 0

Question 13

Complete

Marked out of 1.00

If a dataset has variances: PC1 = 5, PC2 = 3, PC3 = 2, what % variance is explained by PC1?

Select one:

- a. 25%
- b. 50%
- c. 40%
- d. 60%

Question 14

Complete

Marked out of 1.00

Which statement best distinguishes AI software from traditional software?

Select one:

- a. AI avoids all human design
- b. AI can improve from data
- c. AI never uses fixed rules
- d. AI does not need testing

Question 15

Complete

Marked out of 1.00

Which of the following is the best example of AI augmenting human work?

Select one:

- a. AI tutor supporting student learning
- b. AI system replacing all teachers
- c. AI rejecting every weak answer
- d. AI blocking all user questions

Question 16

Complete

Marked out of 1.00

For vectors (1,0) and (0,1), cosine similarity = ?

Select one:

- a. 1
- b. 0
- c. -1
- d. 0.5

Question 17

Complete

Marked out of 1.00

If matrix $A = U\Sigma V^T$, what does Σ represent?

Select one:

- a. Eigenvectors
- b. Singular values
- c. Determinant
- d. Rank

Question 18

Complete

Marked out of 1.00

For matrix:

$$\begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$$

Trace = ?

Select one:

- a. 5
- b. 6
- c. 3
- d. 1

Question 19

Complete

Marked out of 1.00

Which of the following best explains why contextual embeddings are more useful than static word embeddings?

Select one:

- a. They work only on large datasets
- b. They change with sentence context
- c. They remove the need for tokens
- d. They store full grammar rules

Question 20

Complete

Marked out of 1.00

In the 4-layer AI system model, which layer mainly handles raw data from sensors, text, and images?

Select one:

- a. Action
- b. Learning
- c. Perception
- d. Reasoning